Ravinder

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Economics and Planning Unit
Indian Statistical Institute
7, S.J.S. Sansanwal Marg
New Delhi, Delhi 110016

E-mail: ravinder20r@isid.ac.in
X (Twitter): @Ravinder2303
LinkedIn: ravinder2303
Webpage: ravinder_webpage

Visiting Fellowships

- Visiting Scholar, Indira Gandhi Institute of Development Research (IGIDR), Mumbai, India. September-December 2025
- Visiting Ph.D. Fellow, United Nations University World Institute for Development Economics Research (UNU-WIDER), Helsinki, Finland. September-November 2022

Education

- PhD in Quantitative Economics, Indian Statistical Institute, Delhi 2020 - 2026 (Exp.)
(Advisor: Prof. Abhiroop Mukhopadhyay)
- Master in Science in Quantitative Economics (MSQE), Indian Statistical Institute, Delhi 2018 - 2020
- Bachelor of Arts (honors) Economics, Kirori Mal College, Delhi University 2015 - 2018

Research Interests

Human Capital, Development Macroeconomics, Economics of Education, Labour Economics, and Rural Economy.

Working Paper

- **Castelló-Climent, Amparo, Abhiroop Mukhopadhyay, and Ravinder. Transforming rural economies through tertiary education: Evidence from India. No. 2023/89. WIDER Working Paper, 2023.**
Abstract: This paper examines whether tertiary-educated individuals who live in rural India contribute to local prosperity. Using village-level data, we estimate the causal impact of tertiary education on rural development, exploiting the location of Catholic missions circa 1911, historically associated with the expansion of higher education, as an instrumental variable. A one percentage point increase in the village share of tertiary-educated individuals raises per capita consumption by about 7.2 percent, a result robust to using night-light density as an alternative prosperity measure and to a wide range of further checks. Two mechanisms drive this effect: a conventional channel, in which tertiary-educated individuals access productive private-sector employment locally and in nearby towns, and a novel channel, in which they raise agricultural productivity by improving households' access to information, technical advice, and crop diversification. The findings point to a distinct form of structural transformation occurring within rural areas, alongside that driven by migration to cities.

- **Foundational Skills and Labor Market Outcomes: Evidence from India**
  Abstract: This paper estimates the causal effect of foundational skills—basic literacy and numeracy—on adult earnings in India. The central challenge is that no nationally representative dataset links childhood learning to later labor market outcomes. We overcome this by training a machine learning model on the Indian Human Development Survey—which assesses the test scores of children aged 8-11 alongside their rich household characteristics—and then predicting these scores in the Periodic Labour Force Survey, which records the same characteristics and adult labor market outcomes but no test scores. The resulting individual-level predicted test scores, aggregated at the district level, correlate strongly with independent learning data from the Annual Status of Education Report (ASER). For identification, we exploit a fuzzy regression discontinuity design based on the district female literacy threshold that governed eligibility for the District Primary Education Programme. Eligibility raised predicted test scores by about 0.17 standard deviations at the cutoff, and a 0.1 standard-deviation increase in test scores raises earnings by about 8 percent, with robust effects across specifications. Re-estimating the first stage with actual ASER scores confirms that the jump in learning at the cutoff is present in directly measured data, not an artefact of prediction. We also estimate returns using Learning-Adjusted Years of Schooling—which avoids conditioning on schooling years as a bad control and accounts for both the duration and the quality of schooling—and find returns of 19 to 22 percent per 0.1 standard deviation. A potential mechanism operates primarily through occupational sorting: stronger skills raise the probability of regular wage/salaried employment and reduce casual wage work. These findings underscore that labor market success in a developing economy rewards the quality of schooling—learning—not merely its duration.

- **College Expansion, Educational Attainment, and Land Fractionalization: Evidence from Rajasthan, India**
  Abstract: After 2003, Rajasthan relaxed private college entry norms and underwent India's fastest expansion of tertiary education institutions, with the number of colleges rising from 552 in 2003 to 3,393 by 2020 - an 11.3 percent compound annual rate against a 6.2 percent national average. This paper asks where the new colleges were sited, what governed their placement, and whether enlarged physical access raised local educational attainment. Combining six data sources, I examine placement using a random forest over geographic, terrain, demographic, land-use, and infrastructure predictors, and estimate the expansion's effects using border-district event studies and staggered Callaway-Sant'Anna difference-in-differences. Placement tracked towns and existing economic activity rather than human capital, caste composition, or educational need: distance to the nearest town, nighttime light intensity, and geographic location together supply roughly 81 percent of predictive power, while the demographic and equity-relevant characteristics a need-based siting rule would prioritize account for just 4.5 percent. District-level land fractionalization, measured in 1999 before the expansion, strongly predicts subsequent growth in college access, consistent with fragmented, lower-value agricultural land offering cheap sites for a land-intensive, self-financed private-college boom. The expansion raised tertiary attainment among exposed resident cohorts by about 1.8 percentage points and drew education-motivated in-migration of roughly two percentage points. Expanded access, therefore, lifted attainment; but because provision followed pre-existing urban and administrative gradients rather than educational need, and because part of the new capacity was absorbed by mobile students from elsewhere, the expansion traced, rather than redrew, the geography of opportunity it inherited.

Work in Progress

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Master's Thesis

- **Human Capital Inequality and Rural Income - The case of Rural India.**
Abstract: This paper estimates the impact of human capital inequality on rural income for 27 states of India at the village level, taking advantage of Census (2011) and Socio-Economic and Caste Census (2011) data. To address the endogeneity of human capital inequality and mean education, we use the distance to the nearest Catholic and Protestant missionaries (settled in the early 20th century) as an instrument. Our instrumental variable results find a positive effect of human capital inequality and mean education on mean income, while the interaction of these variables shows a negative effect on mean income. The IV results estimate that one standard deviation increase in human capital inequality increases mean income by 221%, and one year increase in mean education increases mean income by 136%. The results are robust to alternative measures of income, such as nightlight intensity. The possible mechanism to explain this relationship is through the occupation structure. Our results are robust and significant at the 1% level.

Other Writing

- **Predicting School Education Outcomes in India: A Machine Learning Approach (with Priya Bareja).**
Abstract: Optimal policy design is a three-step approach —qualitatively capturing various dependencies relating input(s) to output(s), converting these qualitative insights into quantitative relationships that predict the output for various combinations of the inputs and finally, using this quantitative relationship to design optimal input combinations. This paper estimates a prediction model to predict the school learning outcomes for a given set of inputs like teacher characteristics, school infrastructure and health indicators of children. We merge district-level data from four datasets on the National Data and Analytics Platform (NDAP) to get the features and target variables, and train sixteen machine learning algorithms on the data. We use K-fold cross validation and choose the algorithm with the least test error —the Extra Trees Regressor. Thus, learning outcomes produced with a given input combination can be predicted sufficiently accurately by the policymaker.

- **Higher Education and its Impact on the Non-Farm Activities in Rural India (with Abhiroop Mukhopadhyay).**
Abstract: This paper investigates the causal impact of tertiary education on the rural economy of India through entrepreneurship and jobs in the non-farm sector. Leveraging village-level data from the Economic Census 2013 and the Socio-Economic and Caste Census 2011, we construct a measure of the tertiary-educated population and examine its relationship with non-farm economic activity. Our results show that a higher share of tertiary-educated individuals significantly increases the number of non-farm enterprises, entrepreneurs, and overall employment in the rural non-farm sector. Further, it increases the probability of private salaried employment that involve commuting. To address endogeneity concerns, we use the historical presence of Catholic missions as an instrument for tertiary education. The findings contribute to the broader literature on human capital and economic development by highlighting the role of education in shaping the rural economy.

Award, fellowship and Grants

- Best Paper award at 7th Annual Economics Conference, Amrut Mody School of Management, Ahmedabad University. 2026

- Visiting Ph.D. fellowship, IGIDR, Mumbai, India. 2025
- Awarded with an OSF Grant to Attend the XXXI meeting of the Economics of Education Association. 2023
- Project (STEM Education and Economic Growth) Grant by Planning and Policy Research Unit (PPRU). 2023
- Visiting Ph.D. fellowship, UNU-WIDER, Helsinki, Finland. 2022
- First prize (with Priya Bareja) for best research paper and visualization using NDAP data organized by Niti Aayog, Govt of India. 2022
- Senior Research Fellowship, Indian Statistical Institute, Delhi. 2022 - ongoing
- Junior Research Fellowship, Indian Statistical Institute, Delhi. 2020 - 2022
- Masters Fellowship, Indian Statistical Institute, Delhi. 2018 - 2020

Conference and Workshop

- Poster Presentation at WIDER Development Conference, Green Industrialization and Inclusive Growth in a Fractured World Order, ISID, New Delhi. 18-20 March 2026
- Presented at Economics Research Conference (ERC) 2026, Indian Institute of Management Calcutta. 13-14 March 2026
- Presented at 5th Society for Economics Research in India Doctoral Conference (SERI-D) 2026, Amrut Mody School of Management, Ahmedabad University. 06-07 March 2026
- Presented at Research Scholars' Day 2026, Department of Economic Sciences, Indian Institute of Technology Kanpur. 13-14 February 2026
- Presented at 7th Annual Economics Conference, Amrut Mody School of Management, Ahmedabad University. 09-10 January 2026
- Teaching Assistant and Attendee of Workshop on Causal Identification: Rethinking Econometric Methods at Saint Xavier's College, Jaipur. 27-31 December 2025
- Presented at Annual Conference on Economics and Public Policy, O.P. Jindal Global University, Sonipat. 22-23 December 2025
- Teaching Assistant and Attendee of Workshop on Basics of Statistics & Econometric Methods at the Indian Institute of Public Health, Shillong. 24-28 November 2025
- Presented at Symposium on Gender Inequality in Indian Higher Education, Visva-Bharati University, Santiniketan. 04 November 2025
- Presented at 19th Annual Conference on Economic Growth and Development, Indian Statistical Institute Delhi. 19-21 December 2024
- Teaching Assistant and Attendee of Workshop on Data Analytics and Causal Design at the School of Economics at University of Hyderabad. 16-18 August 2024
- Presented at Recent Developments in Economics Research: Theory and Evidence, a conference hosted by Centre for International Trade and Development, Jawaharlal Nehru University, New Delhi. 07-08 March 2024

- Presented at the Meeting of Young Minds in Frontiers of Economics, a conference hosted by INET-YSI at Indian Institute of Technology Bombay. 20-22 February 2024
- Presented at the Winter School hosted by Delhi School of Economics, University of Delhi. 14-16 December 2023
- Presented at the XXXI Economics of Education Association meeting hosted by Santiago de Compostela University. 29-30 June 2023
- Attendee of the RISE Research Conference on Education Economics, Indian Institute of Management Ahmedabad (IIMA). 24-25 February 2023
- Presented at 17th Annual Conference on Economic Growth and Development, Indian Statistical Institute, Delhi. 19-21 December 2022
- Presented in the Advanced Graduate Workshop on Poverty, Development and Globalization, Azim Premji University, Bangalore. 10-23 July 2022
- Teaching Assistant and Attendee of Political Economy Theory and Data Analysis Workshop at GIPE, Pune. 09-11 June 2022

Teaching

- Teaching Assistant for Econometric Method 1 January 2026 - July 2026
- Teaching Assistant for Workshop on Causal Identification: Rethinking Econometric Methods at Saint Xavier's College, Jaipur. 27-31 December 2025
- Teaching Assistant for Workshop on Basics of Statistics & Econometric Methods at the Indian Institute of Public Health, Shillong. 24-28 November 2025
- Teaching Assistant for Econometric Method 1 January 2025 - July 2025
- Teaching Assistant for Workshop on Data Analytics and Causal Design at the School of Economics at University of Hyderabad. 16-18 August 2024
- Teaching Assistant for Econometric Method 1 January 2024 - July 2024
- Teaching Assistant for Econometric Method 1 January 2023 - July 2023
- Teaching Assistant for Pol. Eco. Workshop at Gokhale Institute of Politics and Economics (GIPE), Pune 09-11 June 2022
- Teaching Assistant for Econometric Method 1 January 2022 - July 2022

Professional Activities

Journal Referee

- Economic Modelling

Other Professional Activities

- Discussant at Research Scholars' Day 2026, Department of Economic Sciences, Indian Institute of Technology Kanpur. 13-14 February 2026
- Discussant at the XXXI Economics of Education Association meeting hosted by Santiago de Compostela University. 29-30 June 2023
- Managing the Planning and Policy Research Unit (PPRU), a research center within the Economics and Planning Unit. January 2023 - September 2025
- Association for Mentoring and Inclusion in Economics (AMIE) Mentee. July 2022 - June 2023
- Coordinate weekly seminars at Economics and Planning Unit (with Prof. Debasis Mishra, Prof. E. Somanathan, and Vilok). June 2022 - July 2024

Software Skills

STATA, LaTeX, ArcGIS, Python and R.

References

Professor Abhiroop Mukhopadhyay,
Economics & Planning Unit,
Indian Statistical Institute,
7, SJS Sansanwal Marg,
New Delhi-110016, India.
Email: abhiroop@isid.ac.in
Phone: +91-11-4149-3925

Professor Chetan Ghate,
Economics & Planning Unit,
Indian Statistical Institute,
7, SJS Sansanwal Marg,
New Delhi-110016, India.
Email: cghate@isid.ac.in
Phone: +91-11-4149-3938

Professor S. Chandrasekhar,
Indira Gandhi Institute of Development Research, Gen A K Vaidya Marg,
Goregaon (East), Mumbai 400 065, India.
Email: chandra@igidr.ac.in
Phone: +91-22-6909-6551

Professor Amparo Castelló Climent,
Institute of International Economics,
University of Valencia,
46022 Valencia, Spain.
Email: amparo.castello@uv.es
Phone: +34-96-3828438